

Quiz 1

1. The position function $x(t)$ of a particle moving along an x axis is $x=4.0t-6.0t^2$, with x in meters and t in seconds. At what time does the particle (momentarily) stop, i.e. instantaneous velocity is zero?
2. For the three $\vec{A} = 2.0\hat{i} + 3.0\hat{j} - 4.0\hat{k}$, $\vec{B} = -3.0\hat{i} + 4.0\hat{j} + 2.0\hat{k}$ and $\vec{C} = 2.0\hat{i} - 3.0\hat{j}$ vectors, what is $2\vec{C} \cdot (\vec{A} \times \vec{B})$?